Power for Crossover Trials

2026-04-17

Introduction

In a crossover trial, each subject receives multiple treatments in sequence with a washout period between. The paired comparison cancels between-subject variability, often requiring far fewer subjects than parallel-group trials. The classic 2x2 crossover has two treatments and two periods.

Prerequisites

Paired t-test, power analysis, crossover design basics.

Theory

In a 2x2 crossover, the period-adjusted within-subject difference has variance σ_w^2, typically estimated from prior crossover data. Sample size is analogous to the paired t-test with $\sigma_d = \sigma_w\sqrt{2}$:

$$n \approx 2\sigma_w^2(z_{1-\alpha/2} + z_{1-\beta})^2/\delta^2.$$

Compared to parallel-group with pooled σ: approximately half the subjects for the same δ when $\sigma_w < \sigma_{\text{between}}$.

Carryover effects invalidate the simple formula; a washout period is planned to eliminate them.

Assumptions

- No carryover; washout adequate.
- Period effects modelled if present.
- Within-subject variance known or estimable.

R Implementation

```r
library(PowerTOST); library(pwr)

# 2x2 crossover bioequivalence: CV within = 20%, ratio = 0.95
sampleN.TOST(CV = 0.20, theta0 = 0.95,
             theta1 = 0.80, theta2 = 1.25,
             alpha = 0.05, targetpower = 0.80,
             design = "2x2")

# Therapeutic crossover: mean difference = 5, sigma_w = 10
n_pair <- 2 * (qnorm(0.975) + qnorm(0.80))^2 * 10^2 / 5^2
n_pair
# Equivalent paired-t calculation
pwr.t.test(d = 5 / 10, type = "paired", power = 0.80)
```

Output & Results

Bioequivalence 2x2 at 20 % CV, ratio 0.95: 20 subjects. Therapeutic 5-unit effect with $\sigma_w = 10$: 34 subjects.

Interpretation

"The 2x2 crossover trial requires 20 subjects for 80 % power to establish bioequivalence within the 80-125 % range, assuming a 20 % within-subject CV and a true ratio of 0.95."

Practical Tips

- Crossover saves subjects but doubles measurement commitment per subject; adherence matters.
- Washout must be long enough to eliminate pharmacokinetic and physiological carryover.
- More than two periods (Latin squares, Williams designs) can be more efficient but require careful balancing.
- Exclude subjects with missing period data (or use mixed models).
- For strictly sequential effects (learning, fatigue), random order assignment and period-as-factor in the model help.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation